

Angle-aware conformal certification for oriented object detection in aerial imagery

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Abstract

Oriented (rotated) object detectors in aerial imagery emit point predictions with no distribution-free statement about how far the true box may lie, nor how many objects they silently miss. This paper delivers actionable per-class certificates that close both gaps, validated out-of-sample on DIOR-R across four single-seed detector architectures (training-seed robustness quantified on HRSC2016-MS; multi-seed DIOR-R variance is future work): all 20 classes certified, none refused, with readouts an operator can act on (center-offset radius in pixels, orientation half-extent in degrees, per-class rotated-IoU floor). The certificates rest on a seam-continuous, square-safe Gaussian–Wasserstein (GWD) nonconformity score — porting conformal prediction to oriented boxes is not mechanical, because the box angle is periodic with a $\pm 90^\circ$ seam and degenerate at square boxes. On this score we build two distribution-free guarantees: (G1) per-detection localization coverage given detection, measured out-of-sample over $R=20$ scene splits; and (G2) a Learn-then-Test (Hoeffding–Bentkus) certified false-negative-rate bound that refuses below its power floor. The study spans ten G1 cells over three datasets (DIOR-R, DOTA-v1.0, HRSC2016-MS) and five detector architectures (RTMDet-R, Oriented R-CNN, RoI Transformer, S2A-Net, Oriented RepPoints). On DOTA-v1.0, whose overlapping crops violate exchangeability, G1 sits $\approx 1\text{--}1.2$ points below nominal and is reported as a stress case with the mechanism diagnosed; G2 certifies only where per-class support clears its floor and otherwise refuses. Ships as the open `rotcert` tool with frozen result records.

Keywords: conformal prediction; oriented object detection; uncertainty quantification; distribution-free guarantees; aerial imagery; remote sensing

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1 Introduction

Oriented bounding boxes (OBBs) are the standard output format for detection in aerial, satellite, and document imagery, where objects appear at arbitrary in-plane rotations. Modern oriented detectors (RTMDet-R (Lyu et al., 2022), Oriented R-CNN (Xie et al., 2021), LSKNet (Li et al., 2023)) achieve strong accuracy on benchmarks such as DOTA (Xia et al., 2018) and DIOR-R (Cheng et al., 2022), but a deployed detector emits a point prediction with no distribution-free statement about how far the true box may lie from it, nor about how many objects it silently misses.

Conformal prediction supplies exactly such guarantees, and conformal object detection is an established line: coordinate-wise/corner-wise interval constructions and conformal risk control for axis-aligned detectors (Andéol et al., 2023; Timans et al., 2024; Andéol et al., 2025; Ries et al., 2026; Angelopoulos et al., 2022a), including one aerial (axis-aligned) treatment (Copley et al., 2024). The oriented case is not a mechanical port. An OBB is (cx, cy, w, h, θ) with a long-edge angle θ living on the quotient $\mathbb{R}/\pi\mathbb{Z}$: it is periodic, with a numerical discontinuity at the $\pm 90^\circ$ seam (a true axis at 89° and a prediction at -89° differ physically by 2° but numerically by 178°), and it is unidentifiable at square aspect ratios ($w = h$). A coordinate-wise conformal interval on θ neither wraps nor stays defined at squares, so it is structurally mis-specified in those regimes. We treat this as a structural argument and put it to an empirical, regime-conditional test in §5 rather than asserting the failure as established: we find the naive baseline’s coverage is indeed regime-dependent, though (over-covering globally via Bonferroni) it dips below nominal only in some cells — an honestly scoped result rather than a universal “naive breaks” claim.

Contribution. We conformalize a nonconformity score that is continuous across the angle seam and safe at square boxes — the 2-Wasserstein distance between the Gaussian representations of predicted and true boxes (the GWD, drawn from the oriented-detection loss literature (Yang et al., 2021b,c) but here used as a conformal score, not a loss). On this score we build a two-guarantee certification system: G1, per-detection localization coverage regions with a distribution-free marginal guarantee (Vovk et al., 2005), measured out-of-sample; and G2, a Learn-then-Test (Angelopoulos et al., 2021) Hoeffding–Bentkus certified rotated-IoU false-negative-rate bound with an explicit power floor below which the tool refuses. We report a completed DOTA-v1.0 pilot, a multi-dataset DIOR-R extension (two in-house-trained detectors), a Config-A extension adding an HRSC2016-MS ship arm and a DOTA detector-zoo arm, and a Config-B extension adding cross-family detector breadth to DIOR-R itself (RoI Transformer (Ding et al., 2019), S2A-Net (Han et al., 2022)) — ten G1 cells over three datasets and five detector architectures in all (§5.8, §5.9). Our claimed novelty is narrow and explicit: (i) the angle-aware OBB conformal score and the regime-conditional diagnosis of naive-conformal behaviour; (ii) the two-guarantee oriented-detection certification system; (iii) the `rotcert` tool. We do not claim conformal detection, conformal risk control / Learn-then-Test, the Gaussian OBB representation, or the detectors themselves — these are consumed. Figure 1 summarizes the resulting pipeline: an arbitrary frozen

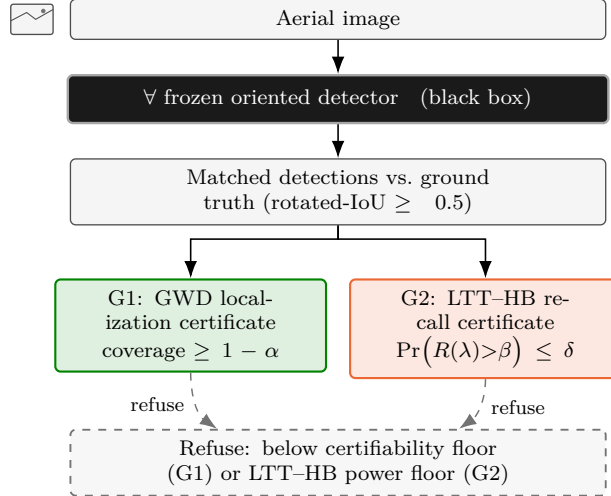


Figure 1: Method overview. The wrapper treats any frozen oriented detector as a black box (five architectures certified in this paper: RTMDet-R-l, Oriented R-CNN, RoI Transformer, S2A-Net, plus the Config-A DOTA detector-zoo arm). Matched detections feed two independent, coupled certificates: G1 Mondrian (per-class) GWD calibration, giving a certified GWD-ball region at coverage $\geq 1 - \alpha$; and G2, a Learn-then-Test Hoeffding–Bentkus (LTT–HB) certified rotated-IoU recall bound. Either branch refuses — rather than reports — on strata that do not clear its own certifiability floor ($\alpha_{\min} = 1/(n_{\text{cal}} + 1) > \alpha$ for G1; below the LTT–HB power floor for G2).

oriented detector is wrapped, never modified, and its matched detections feed two independent certificates — G1 (Mondrian GWD calibration) and G2 (Learn-then-Test certified recall) — either of which refuses on strata below its own certifiability floor rather than reporting a number it cannot back.

2 Related Work

Conformal and risk-controlled object detection. Split conformal prediction (Vovk et al., 2005) and its Mondrian (group-conditional) variant (Vovk et al., 2003) give finite-sample marginal coverage under exchangeability; adaptive, heteroscedasticity-aware conformal regression (Romano et al., 2019) is the broader motivation for conditioning on an informative stratum (class, here) rather than reporting one constant-width interval. For detection specifically, coordinate/corner-wise interval constructions and two-step uncertainty calibration (Andéol et al., 2023; Timans et al., 2024; Ries et al., 2026), and conformal risk control / sequential risk control for recall-type risks (Angelopoulos et al., 2022a; Andéol et al., 2025), target axis-aligned boxes. Learn-then-Test (Angelopoulos et al., 2021) provides high-probability risk certificates via multiple testing over a threshold grid, which we adopt for G2 – itself a multiple-testing instance of the more general risk-controlling-prediction-sets framework (Bates et al., 2021). Distribution-free structured-output UQ has also been built for other per-location targets, notably pixel-wise image-to-image regression (Angelopoulos et al., 2022b), but not, to our knowledge, for the periodic, seam-discontinuous angle parameter of an oriented box. The one aerial treatment we are aware of (Copley et al., 2024) conformalizes an axis-aligned YOLO

detector.

Oriented detection with conformal guarantees (the closest neighbour). The most adjacent system, EAV-DETR (Zuo et al., 2026), couples arbitrary-view oriented UAV detection with a “Pose-Aware Mondrian Conformal Prediction” (PA-MCP) module that uses the UAV flight pose as the Mondrian stratum to produce conditional-coverage prediction sets. Its contribution is a single, bespoke high-accuracy detector: EAV-DETR is a real-time oriented-DETR variant whose gains come from two trained architectural components (scale-adaptive center supervision and anisotropic decoupled rotational attention), evaluated on two UAV benchmarks (CODrone, UAV-ROD), with the conformal layer calibrated on top of that detector and conditioned on a platform-specific pose prior. Our contribution is orthogonal and complementary: a detector-agnostic, purely post-hoc certification wrapper that treats any frozen detector as a black box — five detector architectures across three satellite/aerial benchmarks — strata on object class rather than on capture pose (so it needs no platform metadata), and adds machinery a single built-in guarantee does not carry: explicit certifiability-floor refusals, a coupled certified-recall guarantee via Learn-then-Test (Angelopoulos et al., 2021), and out-of-sample coverage validation. The two lines therefore complement rather than compete — a purpose-built UAV detector with an integrated pose-conditioned set, versus a reusable geometry-certification layer for the frozen detectors practitioners already deploy. (The differentiation here rests on EAV-DETR’s public abstract and code release; the full text is paywalled.)

Uncertainty quantification in the remote-sensing venues. Uncertainty-aware and evidential models are by now an established genre in the remote-sensing literature — e.g. uncertainty-aware building extraction (Li et al., 2024), evidential open-set hyperspectral classification (Ji et al., 2024), and Bayesian uncertainty for explainable SAR target detection (Huang et al., 2023) — but these report heuristic or Bayesian uncertainty scores, not finite-sample coverage guarantees. We instead bring distribution-free, finite-sample conformal and risk-control guarantees to oriented detection in that setting.

Calibration of object detectors. A separate, longer-standing line recalibrates detector confidence scores post hoc, from classifier temperature scaling (Guo et al., 2017) to box-location/scale-conditional calibration for detection specifically (Küppers et al., 2020) and calibration under domain shift (Munir et al., 2022). Calibration reduces the average miscalibration of a score but carries no finite-sample coverage guarantee for any individual stratum; G1 and G2 instead certify — and explicitly refuse below their respective certifiability/power floors rather than report a miscalibrated number.

Oriented detection and the angle representation. DOTA (Xia et al., 2018; Ding et al., 2021) and DIOR-R (Cheng et al., 2022) are the standard oriented benchmarks, with HRSC2016 (Liu et al., 2017) (used here as the elongation-dominated HRSC2016-MS arm, §5.8) and FAIR1M (Sun et al., 2022) the adjacent fine-grained oriented benchmarks we do not certify on. The five architectures our wrapper certifies span the major design lineages of oriented

detection: two-stage RoI-warping (RoI Transformer (Ding et al., 2019), Oriented R-CNN (Xie et al., 2021)), single-stage feature alignment (S2A-Net (Han et al., 2022), RTMDet-R (Lyu et al., 2022)), and point-based representation (Oriented RepPoints (Li et al., 2022), a Config-A zoo detector); the broader lineage these descend from includes the earliest rotated-anchor proposal networks (Ma et al., 2018), vertex-regression (Xu et al., 2021), progressive refinement (Yang et al., 2021a), rotation-equivariant (Han et al., 2021), and attention-based small/cluttered-object (Yang et al., 2019) designs — our wrapper is agnostic to which of these a practitioner deploys, treating each as a frozen black box. All in-house training uses the MMRotate toolbox (Zhou et al., 2022). Gaussian-distance losses (GWD (Yang et al., 2021b), KLD (Yang et al., 2021c)) replace discontinuous angle regression with a smooth distance between box-induced Gaussians; we reuse that representation as a conformal score, not a loss.

3 Methods

The GWD nonconformity score. We canonicalize every ingested box to the 1e90 long-edge convention ($w \geq h$, $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2})$) and represent it as a 2-D Gaussian $\mathcal{N}(\mu, \Sigma)$ with $\mu = (cx, cy)$ and $\Sigma = R(\theta) \text{diag}((w/2)^2, (h/2)^2) R(\theta)^\top$. Σ is a smooth function of (w, h, θ) with no seam discontinuity and no square degeneracy (as $w \rightarrow h$, Σ becomes isotropic and the ill-defined orientation is absorbed). The nonconformity between a prediction $\hat{p}$ and a ground-truth box g is the 2-Wasserstein (GWD) distance

$$s(\hat{p}, g)^2 = \|\hat{\mu} - \mu\|^2 + \text{Tr}\left(\hat{\Sigma} + \Sigma - 2(\hat{\Sigma}^{1/2}\Sigma\hat{\Sigma}^{1/2})^{1/2}\right),$$

with the closed-form 2×2 Bures term; an optional scale normalization $s/\sqrt{\hat{w}\hat{h}}$ is preregistered and ablated.

What the GWD-ball certifies, made actionable. The G1 region is a ball in the GWD metric: $S(\hat{p}) = \{b : s(\hat{p}, b) \leq \hat{q}\}$ certifies, with probability $\geq 1 - \alpha$, that the true box lies within GWD distance $\hat{q}$ of the prediction. Because $s^2 = \|\hat{\mu} - \mu\|^2 + \text{Bures}^2(\hat{\Sigma}, \Sigma)$ is a sum of two non-negative terms, the ball is *contained in* the center slab $\{b : \|\hat{\mu} - \mu\| \leq \hat{q}\}$, so the same certificate implies — with no loss of coverage, by monotonicity of probability under set inclusion — that the true object center lies within $\hat{q}$ pixels of the prediction: a directly usable localization radius (§5.3). GWD is not rotated-IoU: distance in Gaussian-Wasserstein space is monotonically related to, but not a calibrated function of, box overlap, so the ball is not an IoU level-set and must not be read as “the true box is at $\text{IoU} \geq x$ ”; but a user who wants that guarantee obtains it by conformalizing a 1-rotated-IoU score directly, which we promote from an ablation to a reported per-class *certified IoU floor* (§5.3). GWD remains the primary score because it is simultaneously seam-continuous, square-safe, and closed-form; the per-parameter envelope we report for visualization (a bounding box on cx, cy, w, h and an arc on θ) remains a conservative outer approximation and its per-axis widths are not calibrated

marginal intervals.

G1 — per-detection coverage regions, measured out-of-sample. Given matched calibration pairs, let $\hat{q}$ be the $\lceil (1 - \alpha)(n_{\text{cal}} + 1) \rceil / n_{\text{cal}}$ empirical quantile of the scores. We calibrate per class (Mondrian, group = class) and refuse any stratum whose certifiability floor $\alpha_{\min} = 1 / (n_{\text{cal}} + 1)$ exceeds α . Coverage is evaluated out-of-sample: over $R=20$ repeated scene-level splits (calibration 40% / matching 20% / held-out evaluation 40%, split-seed = repeat index), we fit $\hat{q}$ on the calibration scenes and measure coverage on the disjoint evaluation scenes, reporting the across-repeat mean and a BCa 95% CI.

G1 is conditional on detection. It certifies localization given a true positive at the operating rotated-IoU; missed detections and false positives are the job of G2.

G2 — certified rotated-IoU recall / FNR. Retaining detections with confidence $\geq \lambda$, define the per-image risk $R(\lambda) = \text{mean miss rate}$, monotone in λ . We select λ by Learn-then-Test Hoeffding–Bentkus over a λ -grid controlling $\Pr(R(\lambda) > \beta) \leq \delta$. The exchangeable unit is the image, so per-class certification is limited by class-bearing images, not instances. Below the LTT-HB power floor ($n_{\text{img}} \gtrsim \ln(G/\delta) / (2(\beta - \hat{R})^2)$, Bentkus-tightened) the tool refuses and prints the floor.

Baselines and the head-to-head. The same pipeline runs comparison scores through one interface: naive coordinate-wise CP with per-coordinate Bonferroni (B1, the canonical corner-wise conformal object-detection baseline (de Grancey et al., 2022)), axis-aligned-hull CP (B2), and wrapped-geodesic, doubled-angle, 1-rotated-IoU ablations. We report the regime-conditional coverage of B1 in §5 (descriptive). The preregistered efficiency head-to-head (class-blocked bootstrap, Holm-corrected across baseline $\times$ detector $\times$ dataset) ran under the signed freeze and is reported in §5.11, where GWD is smaller in all 8 of 8 cells but is reported as a preregistered descriptive outcome (the single-split test is degenerate; see the disclosure there).

Threats: an in-sample audit defect, found and corrected. An earlier pilot pipeline passed the same matched detection–ground-truth records to both the calibration and audit steps, so the coverage it reported was the split-conformal calibration-set coverage $\lceil (1 - \alpha)(n + 1) \rceil / n$ — for the DOTA pilot, $48,719/54,131 = 0.90002$, which reproduces that in-sample identity exactly and the nominal target to ≈ 4 significant figures (0.9000). That number is an algebraic identity, independent of the score: any nonconformity function (GWD, naive, or noise) yields it, so it has zero power to validate GWD. We correct this by reporting the out-of-sample $R=20$ number throughout (§5); the in-sample value is retained only as a labelled self-consistency diagnostic, never as validation. This is disclosed here rather than left for a reader to infer.

Scene-level discipline. DOTA is tiled into overlapping 1024×1024 crops, so one object appears in adjacent crops; all splitting is at the level of the source scene (never the crop), and every coverage CI is scene-clustered. The

out-of-sample $R=20$ protocol also probes the box-level exchangeability approximation: because each repeat is an independent scene-level partition, the spread of the 20 per-repeat coverages (reported as min/max) is a direct check that the point estimate is stable under scene re-partition, not merely that its CI is wide.

4 Experimental Setup

DOTA-v1.0 pilot. We certify RTMDet-R-l (frozen Apache-2.0 MMRotate (Zhou et al., 2022) zoo checkpoint) on the DOTA-v1.0 (Xia et al., 2018) validation split (15 classes); calibration and evaluation use val only (the test server is untouched). Boxes are canonicalized to $1e90$; matching is greedy one-to-one at rotated-IoU ≥ 0.5 within class. Levels are $\alpha = 0.10$ (G1) and $(\beta, \delta) = (0.20, 0.05)$ (G2). G1 coverage is the out-of-sample $R=20$ scene-split number.

DIOR-R extension. No license-clean DIOR-R-trained checkpoint exists, so both detectors (Oriented R-CNN R-50, RTMDet-R-l) are trained in house on DIOR-R (Cheng et al., 2022) trainval via Apache-2.0 mmrotate; certification runs on the DIOR-R test split (11,738 images; single-tile, scene = image) under the identical protocol. Training setup. Oriented R-CNN is trained $1\times$ (12 epochs) with AdamW ($\text{lr} = 10^{-4}$, weight decay 0.05) rather than the config-default SGD, together with an annotation-filtering step that drops degenerate zero-area DIOR-R ground-truth boxes (removing boxes below a minimum width or height, and images left with no valid ground truth); both changes were made after those boxes destabilized the default recipe. RTMDet-R-l is trained $3\times$ (36 epochs). Checkpoints are saved at the 12-epoch interval with no best-checkpoint hook, so each detector is deployed at the final epoch of its fixed schedule. For RTMDet-R-l the intermediate epoch-24 checkpoint scored higher on the test split (69.65) than the deployed epoch-36 (68.36); because the validation loop evaluated on that same test split, selecting epoch 24 would be checkpoint selection on the evaluation set, so we deploy the final epoch and report the lower number rather than adopt the higher one.

mAP reproduction (AOPG gate, preregistered K3), executed post-freeze. The in-house detectors reproduce DIOR-R test-split mAP of 62.61 (Oriented R-CNN, $1\times$) and 68.36 (RTMDet-R-l, $3\times$), against the AOPG published DIOR-R reference of 64.41 (Cheng et al., 2022). The AOPG table lists neither detector as a same-method row, so its ± 0.5 identity-reproduction tolerance cannot be met by construction ($-1.80, +3.95$ vs. AOPG). A same-method external anchor does exist: a recent benchmark table reports Oriented R-CNN R-50 at 64.30 mAP on DIOR-R (Ding et al., 2026) — essentially AOPG’s 64.41 and within 1.7 points of our $1\times$ reproduction (62.61); the DIOR-R tables that state a schedule use 12 epochs ($1\times$), matching ours. We could not verify any same-method Oriented R-CNN figure near the 67–68 leaderboard band a naive comparison invites: that band is recent end-to-end transformer detectors, not Oriented R-CNN. Both reproduced scores fall in the published DIOR-R band (Oriented R-CNN between Gliding Vertex and RoI Transformer at $1\times$; RTMDet-R-l above the entire 2021 table). We disclose the gap rather than chase it, and certification uses reproduced scores only, never paper-quoted ones.

Table 1: G1 marginal coverage, out-of-sample $R=20$ scene-split (BCa 95% CI; nominal 0.90). The in-sample column is the tautological self-consistency value ($\lceil(1 - \alpha)(n + 1)\rceil/n$), kept as a diagnostic only. From the frozen out-of-sample coverage records.

Cell	detector · dataset	OOS $R=20$ coverage	in-sample (diag.)
DOTA pilot	RTMDet-R-l · DOTA	0.8879 [0.8774, 0.8979]	0.900
DOTA-ORCNN	Oriented R-CNN · DOTA	0.8902 [0.8780, 0.8995]	0.900
DIOR-ORCNN	Oriented R-CNN · DIOR-R	0.9004 [0.8987, 0.9020]	0.900
DIOR-RTMDet	RTMDet-R-l · DIOR-R	0.9012 [0.8995, 0.9030]	0.900

5 Results

5.1 G1 — out-of-sample localization coverage across the four original cells

The GWD certificate’s held-out coverage is at nominal on both DIOR-R cells and $\approx 1\text{--}1.2$ points below on both DOTA cells (Table 1). Decisively, the full detector $\times$ dataset cross now exists: both detector families cover at nominal on DIOR-R (0.9004/0.9012) and below on DOTA (0.8902/0.8879) — the shortfall appears to be a dataset artifact — DOTA’s overlapping 1024-px crops leave residual within-scene correlation that scene-level splitting reduces but does not remove — not a detector or score defect. The in-sample audit (Table 1, right column) is a labelled diagnostic only (see Methods/Threats): it equals 0.900 by construction for any score.

5.2 G1 — per-class Mondrian certificates (all classes certified, none refused)

The Mondrian calibration certifies every class with 0 refused in every cell: 15/15 on DOTA and 20/20 on DIOR-R for both Oriented R-CNN and RTMDet-R-l. Table 2 lists the DOTA pilot’s per-class GWD-ball radii, spanning nearly two orders of magnitude (compact ships/vehicles tightest; elongated ground-track-field/harbor widest); the DIOR-R rosters show the same compact-vs-infrastructure spread (e.g. DIOR ship $\hat{q}=3.99$, area 50 px²; dam $\hat{q}=73.9$, area 17,134 px²).

A recovered-classes provenance note (DIOR-R). The first DIOR-R Oriented R-CNN run certified 18/20 classes: two multi-word classes (expressway-service-area and expressway-toll-station) were excluded because the detector dump lower-cased them while the ground truth capitalized them, so the class-exact matcher found 0 true positives — the tool correctly excluded-and-counted them. Re-running against a casing-normalized ground truth recovers both (20/20) and leaves the other 18 classes’ per-class $\hat{q}$ and n_{cal} bit-identical (verified class-by-class), and out-of-sample coverage unchanged (0.9001 $\rightarrow$ 0.9004) — a surgical fix that also demonstrates the refusal machinery behaving as designed on a real label-hygiene fault.

Per-class conditional coverage on the 20-class dataset (post-hoc, descriptive). A hostile reviewer may ask whether the Mondrian certificate merely holds marginally while hiding per-class shortfalls. It does not. Mirroring the shipped Mondrian machinery (grouping on object class) under the same $R=20$ scene-split protocol, we measure the

Table 2: G1 per-class GWD certificates on the DOTA-v1.0 pilot ($\alpha = 0.10$). $\hat{q}$ is the GWD-ball radius; area is the (cx, cy) coverage-slice area $\pi\hat{q}^2$ (px²). All 15 classes certified, 0 refused. From the frozen per-class certification record.

Class	n_{cal}	$\hat{q}$	Area (px ²)
ship	18,125	3.50	38.6
small-vehicle	10,394	3.14	30.9
large-vehicle	8,727	3.73	43.7
storage-tank	4,291	2.80	24.6
plane	4,299	7.48	175.6
harbor	4,007	19.14	1,151.0
tennis-court	1,510	2.97	27.8
bridge	681	9.34	274.3
swimming-pool	660	6.59	136.3
baseball-diamond	347	10.30	333.1
basketball-court	266	6.37	127.4
roundabout	262	5.55	96.8
soccer-ball-field	238	17.52	963.9
ground-track-field	205	22.09	1,533.1
helicopter	119	12.01	452.9

out-of-sample coverage of the per-class GWD certificate within each of the 20 DIOR-R classes (Table 3). Every class clears the G1 certifiability floor $\alpha_{\min}=1/(n_{\text{cal}}+1) < \alpha$ (0/20 refused; the smallest class, dam, has $n_{\text{cal}}=266$), and per-class conditional coverage is tightly clustered around nominal: pooled per-class coverage is 0.900 in both cells, and no class deviates more than ≈ 1.3 points from 0.90 (Oriented R-CNN range 0.887 golffield – 0.913 dam; RTMDet-R-l range 0.891 stadium – 0.907 tenniscourt), with the sub-nominal classes inside one split-to-split standard deviation of 0.90. The certifiability floor that does bite is the recall certificate’s (G2) images-per-class power floor, not the localization certificate: the small classes it refuses (15/20 for Oriented R-CNN, 14/20 for RTMDet-R-l — last column) are still localization-certified at G1 with coverage at nominal. The geometry certificate thus covers every class approximately at nominal (within ≈ 1.3 points, one split-to-split std, of every sub-nominal class), while the recall certificate honestly abstains on the rare ones.

5.3 G1 — the certificate made actionable: center-offset, orientation, and IoU-floor readouts

A GWD-ball radius is not, on its face, a quantity a remote-sensing practitioner can act on. We therefore translate each frozen per-class $\hat{q}$ — re-derived here from the frozen matched records and gated bit-for-bit against the certification record ($\max |\hat{q} - \hat{q}_{\text{frozen}}| = 0$ across all three cells) — into three interpretable geometric readouts, changing no certified number (Table 5).

(i) *A certified center-offset radius.* Since $s^2 = \|\Delta\mu\|^2 + \text{Bures}^2 \geq \|\Delta\mu\|^2$, the event $\{s \leq \hat{q}\}$ implies $\|\Delta\mu\| \leq \hat{q}$: the GWD-ball lies inside the center slab of half-width $\hat{q}$, so $\Pr(\|\Delta\mu\| \leq \hat{q}) \geq \Pr(s \leq \hat{q}) \geq 1 - \alpha$. The center bound therefore *inherits* the coverage guarantee exactly — it is a consequence, not an approximation. On the DOTA pilot the true center lies within 2.8–3.7 px of the prediction for compact ships and vehicles (up to 22 px for the largest

Table 3: Per-class out-of-sample conditional coverage of the GWD certificate on the two core DIOR-R cells ($\alpha = 0.10$, nominal 0.90; $R=20$ scene splits, mean over repeats). n_{cal} is the Oriented R-CNN full-data per-class calibration count (RTMDet-R-l counts are within a few percent). G2 recall: O=certified for Oriented R-CNN, R=certified for RTMDet-R-l, —=refused by the LTT-HB power floor. Post-hoc / descriptive; from the frozen per-class conditional-coverage record.

Class	n_{cal}	ORCNN cov.	RTMDet cov.	G2 recall
ship	30,731	0.899	0.899	—
storagetank	16,081	0.900	0.900	R
vehicle	12,958	0.904	0.899	—
tennis court	6,498	0.897	0.907	O,R
airplane	5,589	0.898	0.900	—
baseballfield	2,617	0.901	0.898	R
windmill	2,291	0.900	0.895	—
basketballcourt	1,928	0.898	0.894	O,R
groundtrackfield	1,773	0.903	0.905	O,R
harbor	1,692	0.901	0.899	—
bridge	1,470	0.906	0.902	—
overpass	1,183	0.898	0.895	—
expressway-service-area	962	0.896	0.905	O,R
chimney	808	0.899	0.900	—
stadium	600	0.906	0.891	O
expressway-toll-station	508	0.913	0.898	—
golf field	459	0.887	0.899	—
airport	357	0.906	0.896	—
trainstation	354	0.902	0.899	—
dam	266	0.913	0.903	—

infrastructure classes), each with probability ≥ 0.90 .

(ii) *A certified orientation half-extent.* Restricted to the pure-rotation axis through the prediction (fixed center and size), $s^2 = \text{Bures}^2(\Delta\theta)$ is monotone in $|\Delta\theta|$ on $[0, \frac{\pi}{2}]$, so the ball admits rotations up to $\Delta\theta_{\max}$ where $\text{Bures}(\Delta\theta_{\max}) = \hat{q}$ at the class’s median box shape. This resolution is tight exactly where orientation is meaningful (large-vehicle 6.6° , tennis-court 2.9° , ship 11.1°) and correctly *unconstrained* (90°) for the three near-square classes (storage-tank, baseball-diamond, roundabout; aspect ratio ≤ 1.05) whose orientation is genuinely unidentifiable — the square-safety of GWD rendered as an actionable output rather than a caveat. It is an outer envelope (a joint size-and-angle error can also spend the Bures budget), reported as orientation resolution, not a marginal interval.

(iii) *A certified rotated-IoU floor.* Conformalizing 1-rotated-IoU per class with the identical split-conformal estimator yields $\hat{q}_{\text{iou}}$ and hence the directly usable guarantee $\text{rotated-IoU}(\hat{p}, \text{true}) \geq 1 - \hat{q}_{\text{iou}}$ with probability $\geq 1 - \alpha$. Every DOTA class carries a non-vacuous floor between 0.61 and 0.90 (Table 5), and this calibrated floor tracks the empirical 5th-percentile rotated-IoU of the boxes the GWD-ball actually accepts class-by-class (tennis-court 0.902 vs. 0.903; ship 0.757 vs. 0.742), so the abstract GWD ball and the direct IoU certificate agree on the ground. Promoting this construction from the ablation set to a reported certificate answers the practitioner’s “true box at $\text{IoU} \geq y$ ” question without abandoning GWD’s seam-continuity and square-safety for calibration.

Breadth. The pattern replicates on the two Config-B DIOR-R cells (20 classes each): certified center-offset radii

Table 4: The actionable G1 certificate on a second detector family (DOTA-v1.0, Oriented RepPoints, $\alpha = 0.10$; post-freeze addition to the actionable record, same construction and reproduction gate as Table 5). The per-class spread is the point: radii track the calibration support (n_{cal} from 4,355 down to 12), and the two classes with 3–4 TPs are refused as vacuous ($\hat{q} \rightarrow \infty$). 90°: a full flip is within the certified Bures budget — here driven by the radius, not by near-square geometry.

Class	n_{cal}	r_{center} (px)	$\Delta\theta_{\text{max}}$	IoU floor
plane	4,355	13.14	90°	0.719
harbor	631	17.51	26.4°	0.611
baseball-diamond	295	20.19	90°	0.663
swimming-pool	275	8.04	37.8°	0.591
ground-track-field	42	33.75	32.2°	0.736
ship	12	50.90	90°	0.553
soccer-ball-field	4	refused: vacuous (insufficient calibration TPs)		
large-vehicle	3	refused: vacuous (insufficient calibration TPs)		

of 3.5–91 px, non-vacuous IoU floors of 0.53–0.82, and orientation unconstrained on 4–5 near-square classes per cell while tightly bounded on every elongated one (full per-class numbers in the frozen actionable-certificate record).

A second detector family and the per-class spread. The same readouts on the DOTA-v1.0 Oriented RepPoints cell (Table 4) show exactly the per-class differentiation the certificate is meant to expose: plane certifies at 13.1 px center offset with an IoU floor of 0.72 on 4,355 calibration TPs, ground-track-field at 33.7 px / 0.74 on only 42, and ship — on just 12 calibration TPs — at 50.9 px / 0.55, honestly wide rather than silently tight. Two classes with 3–4 calibration TPs are refused as vacuous instead of being reported with unusable radii. The reproduction gate is identical ($\max |\hat{q} - \hat{q}_{\text{frozen}}| = 0$).

5.4 G2 — certified recall scales with images-per-class

At $(\beta, \delta) = (0.20, 0.05)$, G2 certifies a per-class FNR bound where powered and refuses elsewhere (Table 6, all nine cells). The certified count rises from 2/15 on DOTA (~ 458 val images) to 5/20 and 6/20 on DIOR-R (11,738 test images), and RTMDet-R-l (denser detections, more matched TPs) certifies one more class than Oriented R-CNN — direct evidence the LTT-HB power floor is data-scale driven and moves down with more images, not a permanent ceiling. Refusals are of two honest kinds: an infinite power floor (realized per-image miss rate $\geq \beta$, uncertifiable at any n ; e.g. DIOR-ORCNN airplane, vehicle — RTMDet’s floors for the same two classes are finite (1,225 and 75,149 images, $\hat{r} < \beta$), refused only for insufficient images) and insufficient images below the Bentkus floor (e.g. DOTA harbor $n_{\text{img}} = 112 < 135.7$). The tool prints the floor rather than reporting an uncertifiable number.

Table 5: The G1 certificate made actionable on the DOTA-v1.0 pilot ($\alpha = 0.10$). Each frozen per-class GWD radius $\hat{q}$ is read as a certified center-offset radius $r_{\text{center}}=\hat{q}$ (px; inherits the ≥ 0.90 coverage), an orientation half-extent $\Delta\theta_{\text{max}}$ at the median box shape, and a directly-conformalized rotated-IoU floor $1 - \hat{q}_{\text{iou}}$ (≥ 0.90 coverage). †: orientation unconstrained (near-square, aspect ratio ≤ 1.05 ; a full 90° flip is within the certified Bures budget). From the frozen actionable-certificate record; $\hat{q}$ reproduces the certification record exactly.

Class	r_{center} (px)	$\Delta\theta_{\text{max}}$	IoU floor
ship	3.50	11.1°	0.757
small-vehicle	3.14	14.2°	0.708
large-vehicle	3.73	6.6°	0.778
storage-tank	2.80	90° †	0.684
plane	7.48	42.6°	0.808
harbor	19.14	25.0°	0.671
tennis-court	2.97	2.9°	0.902
bridge	9.34	34.9°	0.613
swimming-pool	6.59	31.3°	0.616
baseball-diamond	10.30	90° †	0.726
basketball-court	6.37	13.2°	0.833
roundabout	5.55	90° †	0.753
soccer-ball-field	17.52	24.5°	0.701
ground-track-field	22.09	26.1°	0.791
helicopter	12.01	47.2°	0.686

5.5 G2 made actionable: the power floor is a price tag, not a verdict

The LTT-HB power floor is closed-form in the recorded quantities, $n_{\text{img}} \geq \ln(G/\delta) / (2(\beta - \hat{r})^2 \times 1.75)$, so every refusal can be inverted into an explicit per-class planning readout (Table 7). On the DIOR-ORCNN cell the 15 refusals split into two sharply different regimes. Six classes have a *finite* floor: golffield needs only +261 further images ($491 \rightarrow 752$), and four more classes certify for a one-to-four-fold data increase; every finite-floor refusal also clears at a slightly looser tolerance with the data it already has ($\beta \geq 0.21$ – 0.26 , last column). The remaining nine are *unreachable* at $\beta=0.20$: their realized per-image miss rate already exceeds β ($\hat{r}$ up to 0.502 for dam), so no calibration budget certifies them — the remedy is a better detector or an explicitly looser β , and the tool says which. Table 8 aggregates the same inversion across all cells: at the frozen (β, δ) the cells certify 0–7 classes, while relaxing to $\beta=0.30$ (0.40) would certify 9–14 (10–17) of 20 on the DIOR-R cells — the refusal count is a $(\beta, \text{data-budget})$ dial the practitioner can read directly, reported descriptively.

5.6 Naive coordinate-wise CP under-covers conditionally, not universally

We put the paper’s motivating hook to a direct regime-conditional test (Table 9, descriptive). The naive coordinate-wise (Bonferroni-box) baseline over-covers marginally everywhere (0.933 – 0.936), as Bonferroni predicts, but its coverage is strongly regime-dependent: the seam (boundary) or square is always its most-depressed regime relative to interior. Whether it falls below nominal is cell-dependent — strongly on DOTA square (0.8405 , CI $[0.801, 0.864]$ below nominal; an 11-point deficit vs. its 0.9505 interior), marginally on DIOR-ORCNN boundary (0.8976), and not at all

Table 6: G2 certified-recall counts across all nine cells ($\beta=0.20$, $\delta=0.05$). Certified count scales with images-per-class: the high-image DIOR-R cells certify 5–7/20, while the low-image HRSC2016-MS ($n_{\text{img}}=440$) and DOTA-v1.0 zoo ($n_{\text{img}}=1093$) cells refuse at the LTT-HB power floor. From the per-cell certification outputs.

Cell	cert. / total	certified classes (λ^* , realized risk), or refusal
DOTA pilot	2/15	small-vehicle (0.051, 0.024), tennis-court (0.094, 0.001)
DIOR-R ORCNN	5/20	basketballcourt, tenniscourt, stadium, groundtrackfield, expressway-service-area
DIOR-R RTMDet-R-l	6/20	basketballcourt, tenniscourt, groundtrackfield, expressway-service-area, baseballfield, storagetank
DIOR-R RoI Transf.	5/20	basketballcourt, tenniscourt, stadium, groundtrackfield, expressway-service-area
DIOR-R S2A-Net	7/20	baseballfield, basketballcourt, groundtrackfield, ship, stadium, storagetank, tenniscourt
HRSC ORCNN	0/1	refused: LTT-HB power floor ($n_{\text{img}}=440$, $\hat{r}=0.5464$)
HRSC RTMDet-R	0/1	refused: power floor ($n_{\text{img}}=440$, $\hat{r}=0.6076$)
DOTA RoI Transf.	0/15	refused: power floor ($n_{\text{img}}=1093$, $\hat{r}=0.9380$)
DOTA RepPoints	0/15	refused: power floor ($n_{\text{img}}=1093$, $\hat{r}=0.8496$)

on DIOR-RTMDet (boundary 0.9121, still above nominal). GWD is seam/square-stable on both DIOR-R cells and degrades least at square on the RTMDet-DOTA pilot (≈ 0.88 vs. naive’s 0.8405); on the DOTA-ORCNN cell, however, both scores show a comparable square-regime deficit (GWD 0.8612, naive 0.8591, in-sample diagnostic), so GWD’s square-regime advantage is itself detector-dependent on DOTA rather than universal. We therefore state the hook as a conditional claim, and note that the preregistered set-size head-to-head is reported in §5.11 (GWD smaller in 8/8 cells, reported descriptively); the regime-conditional under-coverage contrast remains descriptive here.

The Config-B DIOR-R extension is regime-consistent for RoI Transformer but not for S2A-Net. Repeating this audit on the two Config-B DIOR-R cells (§5.9) extends the table to all four DIOR-R detectors. RoI Transformer matches the established DIOR-R pattern (boundary 0.902, at nominal, not below it, like DIOR-RTMDet). S2A-Net does not: its boundary coverage (0.867) is the only DIOR-R regime cell in the study that dips clearly below nominal — deeper than DIOR-ORCNN’s marginal dip (0.898) and unlike DIOR-RTMDet or RoI Transformer, both of which sit at or above nominal there — while its square over-coverage (0.988) is the highest of any cell in the table. The direction (boundary worst, square best) is unchanged, but the magnitude is not: the naive baseline’s regime-dependence on DIOR-R is detector-dependent, and S2A-Net is the cell where it is most severe, closer in kind to the DOTA-square deficit than to the mild DIOR boundary dips seen so far.

5.7 τ sensitivity (matched-TP conditioning is not fragile)

Re-matching at rotated-IoU thresholds $\tau \in \{0.5, 0.6, 0.7\}$ (DIOR Oriented R-CNN) tightens the certificate: mean (cx, cy) -slice area $6,514 \rightarrow 4,409 \rightarrow 2,729 \text{ px}^2$ (over the 18 classes powered at all three τ) as τ rises, because higher-IoU true positives have smaller localization error. The realized coverage stays pinned at nominal (0.900) across all

Table 7: Per-class G2 planning readout, DIOR-R Oriented R-CNN ($\beta=0.20$, $\delta=0.05$; post-freeze exploratory, descriptive). Refusals split into finite-floor (reachable with more images, or at a looser β with current data) and infinite-floor ($\hat{r} \geq \beta$: uncertifiable at any budget). From the closed-form inversion of the frozen recall records.

Class	n_{img}	$\hat{r}$	power floor	planning readout
groundtrackfield	—	0.048	met (86)	certified
stadium	—	0.089	met (160)	certified
basketballcourt	—	0.110	met (244)	certified
expressway-service-area	—	0.112	met (253)	certified
tenniscourt	—	0.137	met (491)	certified
golffield	491	0.149	752	+261 img / $\beta \geq 0.21$
windmill	809	0.167	1 762	+953 img / $\beta \geq 0.22$
baseballfield	1 311	0.174	2 852	+1 541 img / $\beta \geq 0.21$
ship	1 404	0.174	2 964	+1 560 img / $\beta \geq 0.21$
storagetank	894	0.177	3 666	+2 772 img / $\beta \geq 0.22$
chimney	447	0.188	14 039	+13 592 img / $\beta \geq 0.26$
airplane	705	0.235	∞	$\beta \geq 0.29$
airport	657	0.461	∞	$\beta \geq 0.52$
bridge	1 299	0.354	∞	$\beta \geq 0.39$
dam	503	0.502	∞	$\beta \geq 0.56$
expressway-toll-station	634	0.238	∞	$\beta \geq 0.29$
harbor	813	0.450	∞	$\beta \geq 0.50$
overpass	1 101	0.277	∞	$\beta \geq 0.32$
trainstation	501	0.305	∞	$\beta \geq 0.37$
vehicle	3 308	0.286	∞	$\beta \geq 0.31$

three τ , but that number is the in-sample calibration-set coverage and is tautologically $\approx$ nominal by the split-conformal construction, so it is a consistency check, not independent evidence. The matched-TP conditioning is thus robust to the match threshold.

5.8 Config-A extension: completing the canonical trio and detector-zoo breadth

Four further G1 cells, run under the identical frozen pipeline ($\alpha = 0.10$, GWD score, scene-level $R=20$ splits), extend the study from two datasets to three and from the two core detectors to four detector architectures (2-stage, dynamic-label, RoI-Transformer, and point-based designs). Two cells add the elongation-dominated HRSC2016-MS ship benchmark (Liu et al., 2017) (the multi-scale variant), certifying two detectors trained in house (Oriented R-CNN 1 $\times$, RTMDet-R 3 $\times$, mirroring the DIOR-R configs); two add detector-zoo breadth on DOTA-v1.0 (RoI Transformer, Oriented RepPoints (Li et al., 2022)). Table 10 reports the audited G1 marginal coverage and the out-of-sample $R=20$ recalibration. The GWD certificate holds at nominal by point estimate in every new cell (audited 0.9003–0.9015; out-of-sample $R=20$ mean 0.888–0.909), though the low-support RoI Transformer cell (70 scenes) carries a wide audit CI ([0.823, 0.945]); the naive coordinate-wise baseline over-covers throughout (audited 0.918–0.933), matching the pattern of the four original cells.

G2 recall: refused in all four new cells — by design. At $(\beta, \delta) = (0.20, 0.05)$ the LTT-HB power floor is infinite

Table 8: G2 planning summary across cells (post-freeze exploratory, descriptive): certified (A), finite-floor refused (B), unreachable refused (C) at the frozen $(\beta, \delta)=(0.20, 0.05)$, and classes certifiable with current data under looser β . From the closed-form inversion of the frozen recall records.

Cell	A/total	B	C	$\beta=0.25$	$\beta=0.30$	$\beta=0.40$
DOTA pilot	2/15	12	1	8	9	10
DIOR-R ORCNN	5/20	6	9	10	13	17
DIOR-R RTMDet-R-l	6/20	8	6	12	12	15
DIOR-R RoI Transf.	5/20	5	10	10	14	17
DIOR-R S2A-Net	7/20	4	9	10	12	13
HRSC ORCNN	0/1	0	1	0	0	0
HRSC RTMDet-R	0/1	0	1	0	0	0
DOTA RoI Transf.	0/15	1	14	0	0	0
DOTA RepPoints	0/15	2	13	0	0	1

Table 9: Angle-regime coverage of the naive coordinate-wise baseline vs. GWD (in-sample audit, nominal 0.90; descriptive). Naive over-covers marginally but dips below nominal only in some cells. Extended to all four DIOR-R cells (RoI Transformer, S2A-Net added, §5.9). From the frozen regime-audit records.

Cell	naive marg.	naive interior	naive boundary	naive square	GWD (int/bnd/sq)
DOTA pilot	0.933	0.951	0.919	0.841	0.893/0.937/0.880
DIOR-ORCNN	0.935	0.944	0.898	0.962	0.886/0.910/0.933
DIOR-RTMDet	0.935	0.945	0.912	0.940	0.885/0.917/0.927
DIOR-RoI-Trans	0.936	0.945	0.902	0.960	0.885/0.910/0.934
DIOR-S2ANet	0.935	0.944	0.867	0.988	0.884/0.911/0.951

in every new cell (the realized per-image miss rate sits at or above β at the available image counts), so G2 refuses rather than reporting an uncertifiable bound: HRSC $n_{\text{img}} = 440$ ($\hat{r} = 0.5464$ Oriented R-CNN, 0.6076 RTMDet-R) and the DOTA zoo subset $n_{\text{img}} = 1093$ ($\hat{r} = 0.9380$ RoI Transformer, 0.8496 Oriented RepPoints). This is the refusal mechanism operating exactly as designed (Methods, G2 power floor; §3), not a failure.

Per-stratum G1 refusals (certifiability floor). In the Oriented RepPoints · DOTA cell (six Mondrian strata) two tiny strata are refused because their certifiability floor $\alpha_{\min} = 1/(n_{\text{cal}} + 1)$ exceeds $\alpha = 0.10$: large-vehicle ($n_{\text{cal}} = 3$, $\alpha_{\min} = 0.25$) and soccer-ball-field ($n_{\text{cal}} = 4$, $\alpha_{\min} = 0.20$). The other three new cells refuse no stratum (HRSC2016-MS has a single ship stratum).

Zoo exclusions (disclosed). We planned six DOTA-v1.0 zoo detectors and realized four: S2A-Net (Han et al., 2022) and Gliding Vertex (Xu et al., 2021) both failed to load (mmrotate v0.1.0-era checkpoints incompatible with the dev-1.x runtime) and are excluded-and-counted rather than silently dropped.

5.9 Config-B extension: cross-family detector breadth on DIOR-R

Two further G1 cells add cross-family detector breadth to DIOR-R itself, run under the identical frozen pipeline as the two original DIOR-R cells (§4): RoI Transformer (two-stage) and S2A-Net (single-stage, dynamic anchor refinement), both trained in house 1× on the same DIOR-R trainval split and certified on the same 11,738-image test split. (S2A-Net’s

Table 10: Extension G1 cells at $\alpha = 0.10$ (nominal 0.90), all under the same frozen pipeline as the core cells (Table 1). Audited marginal coverage (point [CI]) for the GWD score and the naive coordinate-wise baseline, the out-of-sample $R=20$ recalibration (mean [min, max] over 20 scene reseeds), and the matched-true-positive / scene counts. Config-A (§5.8) adds HRSC2016-MS and a DOTA-v1.0 detector zoo; Config-B (§5.9) adds two cross-family DIOR-R cells. From the frozen Config-A and Config-B certification records.

Detector · dataset	GWD coverage	naive-coord coverage	$R=20$ mean [min, max]	n_{TP}	scenes
Config-A — HRSC2016-MS and DOTA-v1.0 detector zoo (§5.8)					
Oriented R-CNN · HRSC2016-MS	0.9011 [0.8798, 0.9201]	0.9212 [0.9000, 0.9418]	0.8989 [0.8528, 0.9500]	799	314
RTMDet-R · HRSC2016-MS	0.9015 [0.8760, 0.9227]	0.9179 [0.8937, 0.9405]	0.9085 [0.8659, 0.9568]	670	310
RoI Transformer · DOTA-v1.0	0.9004 [0.8232, 0.9447]	0.9279 [0.8846, 0.9554]	0.8882 [0.7431, 0.9736]	4,246	70
Oriented RepPoints · DOTA-v1.0	0.9003 [0.8545, 0.9299]	0.9329 [0.9082, 0.9559]	0.8988 [0.7818, 0.9720]	5,617	205
Config-B — cross-family DIOR-R (§5.9)					
RoI Transformer · DIOR-R	0.9000 [0.8961, 0.9036]	0.9364 [0.9340, 0.9388]	0.9014 [0.8919, 0.9113]	89,974	10,966
S2A-Net · DIOR-R	0.9000 [0.8959, 0.9041]	0.9349 [0.9322, 0.9377]	0.9006 [0.8941, 0.9074]	95,192	10,940

native training convention is $1e135$ rather than $1e90$; this has no bearing on the certified score, since every emitted box is re-canonicalized to $1e90$ on output regardless of the model’s internal angle convention, §3.) The GWD certificate again holds at nominal by point estimate on both (audited 0.9000/0.9000; out-of-sample $R=20$ mean 0.9014/0.9006), and the naive coordinate-wise baseline over-covers as usual (0.9364/0.9349 audited) — matching the pattern of the two original DIOR-R cells (Table 10). Both new cells certify every Mondrian stratum at G1 (20/20, 0 refused for GWD and for naive-coord), extending the paper’s “zero G1 refusals on DIOR-R” record to all four detector×DIOR-R cells.

G2 recall and per-class conditional coverage. At $(\beta, \delta) = (0.20, 0.05)$, G2 certifies 5/20 classes for RoI Transformer — the identical roster the Oriented R-CNN cell certifies (basketballcourt, tenniscourt, stadium, groundtrackfield, expressway-service-area, §5) — and 7/20 for S2A-Net (baseballfield, basketballcourt, groundtrackfield, ship, stadium, storagetank, tenniscourt), the highest certified count of any DIOR-R cell, consistent with the paper’s images-per-class scaling account (§5): S2A-Net’s matched-TP population (95,192) is the densest of any cell in the study, and both new cells’ pooled recall refuses only at the LTT-HB power floor ($n_{img} = 19,435$, $\hat{r} = 0.2230/0.2118$), the same refusal mechanism operating exactly as designed elsewhere (§5.8). Repeating the per-class conditional-coverage analysis of §5 (Table 3) on these two cells (now merged into a single four-cell per-class conditional-coverage record) shows the same pattern holds: 0/20 classes refused at the G1 certifiability floor in either cell, and out-of-sample per-class coverage stays inside the paper’s existing ≈ 1.3 -point band around nominal (RoI Transformer range 0.891 groundtrackfield – 0.910 dam; S2A-Net range 0.889 airport – 0.912 trainstation) — the conditional-coverage finding is thus not an artifact of the two original detectors. Figure 6 plots the full four-cell per-class record (all 20 classes across the four DIOR-R detectors), visualizing the tight band around nominal reported above.

Table 11: HRSC 3-seed arm (post-hoc/exploratory Config-B): per-seed certification quantities at $\alpha = 0.10$ (nominal 0.90), GWD score, same frozen pipeline. Audited marginal coverage (point [CI]), matched true-positive count, out-of-sample $R=20$ recalibration (mean [min, max] over 20 scene reseeds), and G2 refusal status. Seed 0 is the already-certified Config-A cell (not recomputed); seeds 1, 2 are added here.

Detector	seed	audited GWD coverage [CI]	n_{TP}	$R=20$ mean [min, max]	G2
Oriented R-CNN	0	0.9011 [0.8798, 0.9201]	799	0.8989 [0.8528, 0.9500]	refused (power floor)
Oriented R-CNN	1	0.9023 [0.8803, 0.9228]	727	0.8982 [0.8414, 0.9360]	refused (power floor)
Oriented R-CNN	2	0.9024 [0.8780, 0.9246]	758	0.9005 [0.8249, 0.9545]	refused (power floor)
RTMDet-R	0	0.9015 [0.8760, 0.9227]	670	0.9085 [0.8659, 0.9568]	refused (power floor)
RTMDet-R	1	0.9016 [0.8799, 0.9235]	630	0.9113 [0.8839, 0.9402]	refused (power floor)
RTMDet-R	2	0.9023 [0.8796, 0.9232]	737	0.8903 [0.8444, 0.9311]	refused (power floor)

5.10 Training-seed stability of the certificates (HRSC 3-seed arm)

As a post-hoc, exploratory Config-B arm (not a preregistered confirmatory family) we retrained the two in-house HRSC2016-MS detectors under two further training seeds (seeds 1, 2; seed 0 is the already-certified Config-A cell above, not recomputed) and re-ran the frozen certification chain on the pulled detections. Table 11 reports, per seed, the audited GWD marginal coverage, the matched-true-positive count, the out-of-sample $R=20$ recalibration, and the G2 refusal status.

Three observations, each scoped to what the arm supports. (a) Coverage lands at nominal for every seed — as split-conformal calibration guarantees under exchangeability, not as a nontrivial stability finding: the audited GWD coverage sits within 0.0013 of a common value across seeds (Oriented R-CNN 0.9011–0.9024; RTMDet-R 0.9015–0.9023), i.e. no seed-driven exchangeability breakdown occurred. Split conformal pins the marginal coverage by construction, so this confirms the pipeline rather than discovering a nontrivial stability. (b) The seed-sensitive quantity is the region scale. The conformal quantile $\hat{q}$ (the GWD nonconformity radius, in Gaussian–Wasserstein score units) does vary with the training seed, and unevenly across detectors: the per-seed mean $\hat{q}$ (over the $R=20$ reseeds) moves only 1.5% for Oriented R-CNN (39.57–40.17) but $\approx 12\%$ for RTMDet-R (48.23–54.26), so it is the certified region’s size, not its coverage, that carries the training-seed variation. The six out-of-sample $R=20$ coverage means all straddle nominal (0.8903–0.9113), with RTMDet-R’s wider across-seed spread (0.021) inside each cell’s own reshuffle range (per-cell [min, max] spans ≈ 0.06 –0.13). (c) Refusal is seed-stable: the G2 LTT-HB power-floor refusal ($n_{img} = 440$, below the floor) fires identically in all six cells, so refusal behavior does not flicker with training randomness. The matched-true-positive counts vary with seed (630–799) because the detection sets differ — the expected detector-side variation the certification wraps, not a pipeline instability. (Multi-seed DIOR-R detector-depth cells are left to future work.)

5.11 Preregistered head-to-head: the Holm-8 family (demoted to descriptive)

Preregistered result (descriptive). The preregistered family — 2 baseline contrasts (naive coordinate-wise + Bonferroni; axis-aligned hull) $\times$ 2 detectors $\times$ 2 datasets = 8 one-sided tests (H1: GWD region strictly smaller) — was executed exactly as frozen under the signed freeze on a single frozen 40/20/40 cal/match/eval scene split (split-seed 0) per cell; the $R=20$ across-split robustness below is a separate, exploratory analysis. GWD is strictly smaller in all 8 of 8 cells. We report this as a preregistered descriptive outcome, not statistical confirmation: as detailed next the single-split test is structurally degenerate, so the nominal Holm-adjusted $p = 0.004$ is an artifact of the $1/2001$ permutation floor and carries no evidence strength. The eval-split region-size ratios (GWD $\div$ baseline; “region size” here = the (cx, cy) -slice area, $\pi\hat{q}^2$ for GWD and $4\hat{q}_{cx}\hat{q}_{cy}$ for the union-bound baselines — a coverage-region slice, not a calibrated marginal interval; see §3) of 0.67/0.86 (RTMDet–DOTA, vs. naive/hull), 0.60/0.78 (ORCNN–DOTA), 0.33/0.48 (ORCNN–DIOR-R), and 0.40/0.59 (RTMDet–DIOR-R): GWD is strictly smaller in every cell.

Degeneracy disclosure — the p -values carry no evidence strength. Under the frozen protocol both constructions are calibrated with pooled split-conformal (one marginal cell, no Mondrian grouping) — not the per-class Mondrian calibration of the shipped G1 certificate (§5), a deliberate mismatch we flag here. (The contrast also pooled all matched true positives without excluding square-stratum detections; the design’s square-vacuity rule guards the angular set, not this center-localization (cx, cy) slice, so the effect on the area metric here is limited.) Pooled calibration makes each construction’s region-size a single scalar per cell: $\pi\hat{q}^2$ and $4\hat{q}_{cx}\hat{q}_{cy}$ are functions of scalar quantiles only, so every detection in a cell carries the identical set size (we re-verified 1 unique value over all 20k–40k eval pairs per cell; e.g. ORCNN–DOTA GWD 126.6, naive 212.5, hull 162.3 px²). The preregistered per-pair machinery is therefore degenerate: every per-pair log-ratio equals the same negative constant, so the class-blocked bootstrap CI is zero-width and the sign-flip permutation p is deterministic at the $1/(n_{\text{perm}}+1) = 1/2001$ floor for any negative effect, independent of magnitude; the effective sample size is 1 (a pair of scalars from one split), not the 20k–40k pairs. The preregistered test thus certifies only the per-split deterministic inequality $\pi\hat{q}_{\text{GWD}}^2 < 4\hat{q}_{cx}\hat{q}_{cy}$ on the frozen split; its substantive content is the effect size (the ratios above), not the p -values, and “8/8 at the p -floor” is not eight independent strong rejections. This inequality is moreover close to analytic: the union-bound (Bonferroni) baselines over-cover by construction at any fixed nominal α , so at matched nominal level their (cx, cy) box is predictably larger than a near-nominal ball — the “8/8 smaller” direction largely reflects that structural Bonferroni looseness, not a data-driven margin. The coverage-matched ablation below is precisely what removes it.

Across-split sensitivity of the nominal-level contrast ($R=20$; not preregistered). We recompute the per-split GWD and baseline region sizes on the same $R=20$ scene splits as the G1 coverage headline (split-seed = repeat index), to check how the frozen single-split nominal-level ratio varies by realization. The direction is stable: GWD’s region is smaller on all 20 of 20 splits in every one of the 8 cells, no split’s ratio crosses 1, and the worst single split anywhere is

Table 12: Across-split sensitivity ($R=20$, not the preregistered Holm-8) of the GWD-vs-baseline region-size ratio at matched nominal α ($\text{GWD} \div \text{baseline}$; <1 favors GWD). Per cell: mean and [min, max] over 20 scene splits (split-seed = repeat index, same protocol as the G1 headline) and the count of splits with GWD smaller. The 20 splits reshuffle one fixed scene set (dependent), so no p -value is attached; every split favors GWD and no ratio crosses 1. These nominal-level ratios are re-assessed on a coverage-fair footing in Table 13.

Cell	Baseline	mean ratio	[min, max]	GWD smaller
RTMDet–DOTA	naive	0.597	[0.519, 0.791]	20/20
RTMDet–DOTA	hull	0.773	[0.687, 0.974]	20/20
ORCNN–DOTA	naive	0.573	[0.468, 0.683]	20/20
ORCNN–DOTA	hull	0.746	[0.635, 0.876]	20/20
ORCNN–DIOR-R	naive	0.342	[0.307, 0.395]	20/20
ORCNN–DIOR-R	hull	0.493	[0.460, 0.567]	20/20
RTMDet–DIOR-R	naive	0.381	[0.358, 0.431]	20/20
RTMDet–DIOR-R	hull	0.549	[0.505, 0.625]	20/20

0.974 (RTMDet–DOTA vs. hull). Table 12 reports per-cell mean and [min, max] ratios. We attach no p -value to this 20/20 count: the split procedure reshuffles one fixed scene set, so the 20 splits are dependent, overlapping partitions, not 20 independent draws — an exact sign test would treat them as independent and materially overstate the evidence. We therefore read this only as a split-sensitivity range of the (still descriptive) frozen nominal-level contrast, which it corroborates and nowhere contradicts. Crucially, this analysis (like the frozen family) compares at matched nominal α , where the baselines over-cover; the coverage-matched ablation next is what tests the efficiency claim on a coverage-fair footing, and is our primary quantitative support for it.

Coverage-matched ablation — the efficiency claim on a coverage-fair footing. The contrasts above equalize the nominal $\alpha = 0.10$, not realized coverage. At that level GWD attains close-to-nominal eval coverage ($R=20$ mean 0.883 on DOTA and 0.899 on DIOR-R, consistent with the G1 headline), while the union-bound baselines over-cover — $R=20$ mean realized coverage 0.915–0.948 across cells — so part of the nominal size gap is the price of baseline over-coverage. We therefore run the coverage-matched ablation the previous draft deferred: holding GWD at $\alpha = 0.10$, we re-tune each baseline to a level α' at which its realized eval coverage matches GWD’s on the same split, then compare (cx, cy) -slice areas (Table 13; frozen split and $R=20$ range, realized coverage matched to within ≤ 0.0002). Most of the nominal advantage does not survive. Against the like-for-like naive Bonferroni baseline — which, like GWD, certifies the full 5-DOF oriented box — GWD stays smaller on DIOR-R across all four cross-family detectors now certified there (§5.9): $\approx 0.79\times$ ORCNN, $\approx 0.83\times$ RTMDet, $\approx 0.76\times$ RoI Transformer, $\approx 0.85\times$ S2A-Net (range 0.76–0.85 $\times$; smaller on all 20/20 splits in every one of the four cells), but on DOTA it is a wash (mean ≈ 0.93 , only 12/20 splits favor GWD, the range spanning 1). Against the axis-aligned hull, GWD is larger at matched coverage in every cell run under this ablation, now six in total (1.46–1.76 $\times$ the four original cells; 1.66 $\times$ S2A-Net and 1.75 $\times$ RoI Transformer, both within the same band): the nominal hull advantage was over-coverage plus the hull certifying only the coarser 4-DOF axis-aligned envelope (it drops the angle), so its center footprint is tighter at matched marginal coverage — a

Table 13: Coverage-matched efficiency ablation (post-hoc, coverage-fair; not preregistered). Region-size ratio (GWD $\div$ baseline; <1 favors GWD) with each baseline re-tuned to GWD’s realized eval coverage. Columns: the nominal- α ratio (Table 12 for the four original cells; for the two Config-B DIOR-R cells, not part of that preregistered family, this ablation’s own $R=20$ nominal- α mean) and the coverage-matched ratio ($R=20$ mean [min, max]), plus splits with GWD smaller. The naive baseline certifies the full oriented box (like GWD); the hull certifies only the coarser axis-aligned envelope. From the coverage-matched records.

Cell	Baseline	nominal	matched [min, max]	GWD smaller
RTMDet-DOTA	naive	0.60	0.93 [0.47, 1.26]	12/20
RTMDet-DOTA	hull	0.77	1.48 [1.36, 1.63]	0/20
ORCNN-DOTA	naive	0.57	0.93 [0.42, 1.23]	12/20
ORCNN-DOTA	hull	0.75	1.46 [1.34, 1.61]	0/20
ORCNN-DIOR-R	naive	0.34	0.79 [0.67, 0.89]	20/20
ORCNN-DIOR-R	hull	0.49	1.76 [1.65, 1.81]	0/20
RTMDet-DIOR-R	naive	0.38	0.83 [0.74, 0.92]	20/20
RTMDet-DIOR-R	hull	0.55	1.70 [1.63, 1.82]	0/20
RoI-Trans-DIOR-R	naive	0.33	0.76 [0.62, 0.86]	20/20
RoI-Trans-DIOR-R	hull	0.47	1.75 [1.67, 1.81]	0/20
S2A-Net-DIOR-R	naive	0.38	0.85 [0.74, 0.97]	20/20
S2A-Net-DIOR-R	hull	0.54	1.66 [1.56, 1.72]	0/20

representational caveat, not a GWD defect. At the pooled level the coverage-fair advantage is narrow: GWD is $\approx 0.8\times$ the like-for-like naive oriented-box region on DIOR-R (all four detectors, every split), a wash on DOTA, and larger than the hull everywhere; but this pooled view averages over object shape, and conditioning on it (next, Table 14; the regime-conditional analysis below is not re-run on the two Config-B DIOR-R cells) shows that, within each of the four frozen cells, the naive-baseline advantage is larger on the more elongated objects — a descriptive within-cell pattern, not a mechanism. This ablation is post-hoc but coverage-fair, and is not confirmatory (no preregistration claim). The premise-death branch of C2 did not fire, and the paper’s correctness contributions (seam-continuity, square-safety, the coupled G1/G2 guarantees) do not rest on the efficiency margin. Figure 7 plots the per-cell coverage-matched ratio against both baselines directly, with the $R=20$ range and parity (ratio = 1) marked.

Within each frozen cell, the coverage-fair naive-baseline advantage is larger on more elongated objects (descriptive). The pooled ratios above average over object shape, which hides the mechanism. Repeating the coverage-matched comparison conditioned on the matched ground-truth box’s aspect ratio $AR = \max(w, h)/\min(w, h)$ — a property of the ground-truth object, computed without reference to the certified region, so the strata are not circular, and reusing the parent’s per-split calibration verbatim (the unconditioned “all” regime reproduces Table 13 bit-for-bit; regression-guarded) — resolves the pooled numbers into a monotone within-cell trend across aspect-ratio strata (Table 14). Against the like-for-like naive per-coordinate baseline, at realized coverage matched to ≤ 0.0004 in every regime, GWD’s (cx, cy) region is smaller for elongated objects ($AR \geq 3$) in all four original cells ($0.74\text{--}0.87\times$), while for compact objects ($AR < 3$) it is a wash-to-slight-loss ($0.97\text{--}1.08\times$, the slight loss confined to DOTA). A per-cell median split (balanced halves, cut $\approx 2.2\text{--}2.5$) gives the same direction in all four original cells and sharpens it on DIOR-R, where the more-elongated half reaches $0.59\text{--}0.65\times$ against $\approx 1.02\times$ on the compact half. This explains

the DOTA “wash” (≈ 0.93) rather than explaining it away: it is the average of a genuine elongated-object win and a slight compact-object loss, not evidence that the angle-aware region is never sharper — within these four frozen cells the efficiency advantage is concentrated on the more elongated objects. We report this as a description of these four cells, not as a mechanism: HRSC2016-MS, an all-elongated population (eval frac AR ≥ 3 0.90/0.93), landed at pooled coverage-matched ratios 1.2413/1.0198 — outside the four cells’ elongated band (0.74–0.87 $\times$) and in the wrong direction — so the within-cell pattern is descriptive only and did not survive its one out-of-sample test (§5.8).

Two scope limits bound the claim honestly. First, this is specifically a repair versus the naive per-coordinate baseline: against the axis-aligned hull GWD is larger in every regime (1.4–1.8 $\times$) with no elongation gradient (elongated $\geq$ compact in 3/4 cells), so the advantage is not “smaller than the hull.” Second, the between-dataset pattern is not an elongation effect: DOTA objects are on average more elongated than DIOR-R (mean AR 2.70 vs. 2.35), yet GWD wins pooled on DIOR-R and only ties on DOTA, so the DIOR-vs-DOTA gap tracks the DOTA crop/exchangeability artifact (§5), not shape — the elongation effect is within-cell and conditional. Finally, we can now report the outcome of a registered prediction (logged 2026-07-13, before the Config-A box run) for the staged third-dataset arm: because HRSC2016-MS ships (single-class; source mean AR 5.22, 82% at AR ≥ 3) lie almost entirely in the elongated regime, their pooled coverage-matched ratio versus naive-coord was predicted to be the most decisive GWD win of any cell (low-0.7 $\times$ or below); a wash there would falsify the elongation mechanism. The prediction is reversed. The realized HRSC pooled ratios are 1.2413 (Oriented R-CNN) and 1.0198 (RTMDet-R) — both ≥ 1.0 (mean 1.1305): GWD’s matched-coverage region is larger than naive-coord’s on the most elongated dataset. The two DOTA zoo cells on the same dataset and eval set add strong detector-dependence in the same direction (1.6618 RoI Transformer vs. 0.7056 Oriented RepPoints). We therefore withdraw any cross-dataset elongation-mechanism claim: the elongation effect does not transfer between datasets or detectors. What survives is the within-cell elongation pattern on the four original frozen cells (Table 14), reported as a descriptive, post-hoc observation only; and, as on every earlier cell, GWD never beats the axis-aligned hull (the new cells give 1.30–2.44 $\times$: HRSC 1.82/1.30, zoo 2.44/1.43, consistent with the frozen 1.4–1.8 $\times$). Two population caveats bound this cross-cell reading: HRSC2016-MS has essentially no compact stratum (eval mean AR 5.63/5.95, frac AR ≥ 3 0.90/0.93; its compact regime rows are skipped for want of a powered split), and the DOTA zoo cells run on a 1093-image subset whose matched population is far less elongated (mean AR 1.27/1.47, frac AR ≥ 3 0/0.039) than the frozen DOTA cells (mean AR ≈ 2.7 , frac ≈ 0.29 –0.30), so the zoo ratios are detector-breadth evidence, not a third elongation datapoint.

Confinement of the coverage-fair efficiency claim. Netted out, the coverage-fair naive-baseline size advantage is currently demonstrated on one of the three datasets: GWD is ≈ 0.76 –0.85 $\times$ the like-for-like naive region on DIOR-R (all four detectors — ORCNN, RTMDet, RoI Transformer, S2A-Net — all 20/20 splits), a wash on DOTA (≈ 0.93), and reversed on HRSC2016-MS (1.2413/1.0198, both ≥ 1.0). The DOTA wash was attributed to a crop/exchangeability

Table 14: Regime-conditional coverage-matched efficiency versus the naive per-coordinate Bonferroni baseline (post-hoc, coverage-fair; not preregistered). Ratio $\text{GWD} \div \text{naive}$ of the (cx, cy) -slice area (<1 favors GWD), conditioned on the matched ground-truth box aspect ratio $\text{AR} = \max(w, h)/\min(w, h)$; realized coverage matched to ≤ 0.0004 in every regime. For the four original frozen cells the “all” column reproduces Table 13’s pooled ratio bit-for-bit (regression-guarded), and elongated objects favor GWD while compact objects are a wash-to-slight-loss. The out-of-sample Config-A extension cells (HRSC2016-MS, DOTA-v1.0 zoo; §5.8) do not transfer the gradient — HRSC is ≥ 1 despite being almost all elongated, and the DOTA-zoo cells are near-all compact (—=regime with no powered split). Against the axis-aligned hull the gradient does not hold (text). From the regime-conditional coverage-matched records.

Cell	AR mean (frac ≥ 3)	all	compact (AR <3)	elongated (AR ≥ 3)
Four original frozen cells (within-cell elongation gradient)				
RTMDet-DOTA	2.69 (0.29)	0.93	1.03	0.85
ORCNN-DOTA	2.71 (0.30)	0.93	1.08	0.74
ORCNN-DIOR-R	2.38 (0.22)	0.79	0.98	0.87
RTMDet-DIOR-R	2.34 (0.21)	0.83	0.97	0.86
Out-of-sample Config-A extension (gradient does not transfer)				
HRSC ORCNN	5.63 (0.90)	1.24	—	1.26
HRSC RTMDet-R	5.95 (0.93)	1.02	—	0.98
DOTA RoI Transf.	1.27 (0.00)	1.66	1.66	—
DOTA RepPoints	1.47 (0.04)	0.71	0.74	—

artifact of DOTA’s overlapping tiles (§5, §7), but that account cannot rescue the claim’s generality: HRSC2016-MS is certified single-tile (scene = image, uncropped, 453 whole test images resized in place), so its reversal on the most elongated dataset cannot be a crop artifact. The efficiency advantage is therefore reported as DIOR-R-specific, not general.

6 Discussion

The study is best read as a certification layer for oriented object detectors rather than a new detector architecture. Pattern Recognition reviewers should therefore judge the contribution by whether the layer turns fixed detector outputs into actionable, distribution-free readouts: a center-radius bound, an orientation-extent bound, a rotated-IoU floor, and a separate recall certificate with explicit power-floor refusals. The strongest evidence for that claim is the out-of-sample $R=20$ recalibration record: the DIOR-R class certificates survive across detector families, while DOTA and G2 expose the calibration-budget and exchangeability limits instead of being hidden behind a single pooled score.

This separation also explains why negative results are retained as first-class outputs. The DOTA exchangeability warning, the G2 refusals for rare classes, and the degenerate coverage-matched efficiency test are not failures of the submission package; they delimit when the certificates are informative. In a deployment setting, the practical rule is to publish the certificate only for cells whose calibration pool can support the target risk and to report refusal otherwise. The same logic motivates the full-study design below: the present evidence is enough to support the per-class certification interface and its DIOR-R behavior, but cross-dataset claims about angle-aware efficiency require the staged

extension rather than extrapolation from the frozen pilot.

7 Limitations

Certificate scope (v1). G1 is conditional on detection: it certifies localization for matched true positives, not whether an object is detected (G2’s separate guarantee). The certified object is the GWD-ball in Gaussian-Wasserstein space, not an IoU level-set (Methods); its per-parameter envelope over-covers and is not a set of calibrated coordinate intervals. Guarantees are valid for the calibration distribution only; the tool prints a recalibrate-on-drift caveat.

Scale and the DOTA under-coverage. The out-of-sample DOTA coverage (0.888) sits ≈ 1.2 points below nominal; we attribute this to residual within-scene correlation in DOTA’s overlapping crops (the same detector covers at nominal on single-tile DIOR-R), but a weighted/adaptive conformal remedy for the crop structure is future work. The $R=20$ per-split spread bounds how much this varies by realization.

Frozen-checkpoint variance. The DOTA detector is a frozen published checkpoint (no training-seed variance; variance is over split repeats). The DIOR-R and HRSC2016-MS detectors are trained in house; for HRSC2016-MS a post-hoc 3-seed arm (§5.10) quantifies training-seed variance: audited coverage is seed-stable by construction (within-detector spread ≤ 0.0013) while the certified region scale $\hat{q}$ does vary with seed (up to $\approx 12\%$ for RTMDet-R), and all four DIOR-R cells remain single-seed (multi-seed DIOR-R detector variance is future work). The two DOTA zoo detectors (RoI Transformer, Oriented RepPoints, §5.8) are frozen published checkpoints; note that “RoI Transformer” therefore names two distinct trained instances in this paper — a frozen DOTA-v1.0 zoo checkpoint (§5.8) and an independently in-house-trained DIOR-R model (§5.9) — never the same weights certified twice.

Statistical significance and scoop currency. The regime-conditional under-coverage contrast is descriptive here; the preregistered Holm set-size head-to-head ran under the signed freeze and is reported in §5.11, where the frozen single-split family is demoted to a preregistered descriptive outcome (GWD smaller in 8/8 cells at nominal α) — structurally degenerate, so not presented as confirmatory and not amended. Because that 8/8 is at matched nominal α , where the baselines over-cover, our primary quantitative characterization of efficiency is the post-hoc, coverage-fair coverage-matched ablation (§5.11, Tables 13–14): it shows the size advantage is largely baseline over-coverage, and that the residual advantage over the like-for-like naive baseline is, descriptively and within each of the four original cells, larger on more elongated objects — GWD’s region is smaller for elongated objects (aspect ratio ≥ 3) in all four original cells (0.74–0.87 $\times$) and a wash-to-slight-loss for compact ones — while against the axis-aligned hull GWD is larger in every regime. This gradient is now scoped to a within-cell, descriptive observation: the falsifiable prediction for the elongation-dominated HRSC2016-MS ship arm was executed and reversed (§5.8, §5.11; HRSC pooled ratios 1.2413/1.0198, both ≥ 1.0), so the elongation mechanism does not transfer across datasets or detectors and no

cross-dataset efficiency claim is made. The $R=20$ across-split analysis is reported as split-sensitivity of the nominal contrast, with no p -value (the 20 splits reshuffle one fixed scene set and are not independent). The differentiation against EAV-DETR (Zuo et al., 2026) (§2) rests on its public abstract and code release, its full text being paywalled. The DIOR-R AOPG mAP reproduction gate was executed (reproduced test mAP 62.61 / 68.36 vs. AOPG 64.41; ± 0.5 not met cross-method; same-method external anchor 64.30 (Ding et al., 2026), disclosed).

8 Full-study design

The pilot and DIOR-R cells instantiate the G1/G2 arms; the preregistered study adds the $R=20$ repeats for every cell (done for the coverage headline) and the preregistered Holm family (the paper’s only Holm family: 2 baseline contrasts $\times$ 2 detectors $\times$ 2 datasets = 8 tests), which ran under the signed freeze and is reported in §5.11 (GWD smaller in 8/8 cells, demoted to a preregistered descriptive outcome; no amendment). The AOPG mAP reproduction gate has now been executed (§5); no preregistration-gated slot remains. The staged third-dataset (HRSC2016-MS) arm and the DOTA detector-zoo breadth were executed as the Config-A extension (§5.8); the registered elongation prediction it tested is reported reversed (§5.11). The cross-family DIOR-R detector-breadth arm (RoI Transformer, S2A-Net) was executed as the Config-B extension (§5.9); both new cells certify at G1 with zero Mondrian refusals.

9 Conclusions

This paper set out to give oriented object detection in aerial imagery a distribution-free reliability layer, and delivers it as actionable per-class certificates: validated out-of-sample on DIOR-R across four detector architectures with all 20 classes certified and none refused, and translated into readouts an operator can act on — a certified center-offset radius in pixels, an orientation half-extent in degrees that is tight exactly where orientation is meaningful, and a per-class rotated-IoU floor. The enabling observation is methodological: the 2-Wasserstein distance between box-induced Gaussians is a seam-continuous, square-safe nonconformity score, while a coordinate-wise angle interval is structurally mis-specified at the $\pm 90^\circ$ seam and at square boxes (Figure 2). Two companion guarantees bound what the main certificate does not cover: a Learn-then-Test certified recall bound (G2), and explicit refusal semantics wherever a stratum’s calibration pool cannot support the requested level.

The study’s weaknesses are equally concrete. On DOTA-v1.0 the certificate sits 1–1.2 points below nominal because the standard overlapping-crop pipeline correlates neighbouring evaluations — the guarantee’s exchangeability premise, not the score, is what breaks; a weighted or scene-adaptive conformal construction for crop structure is the clear next step. G2 certifies only classes with large image counts and refuses most classes at realistic per-class support. The efficiency

advantage of the GWD region over naive baselines is established only descriptively — the preregistered inferential family is structurally degenerate — and does not transfer across datasets (a registered elongation prediction reversed on the HRSC2016-MS arm). Finally, all DIOR-R detectors are single-seed, and the certified detectors reproduce below their published mAP.

Beyond the crop-structure remedy, future work includes multi-seed detector variance on DIOR-R, drift-triggered recalibration (the tool already prints a recalibrate-on-drift caveat), and extending the same certification layer from boxes to instance-segmentation masks. The `rotcert` tool, frozen result records, and golden-file-tested CLI accompany the paper.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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CRedit authorship contribution statement

Zeyu Fu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) in order to assist with code development, experiment orchestration, and language editing. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Data and Code Availability

DOTA-v1.0 (public; val GT used, test server untouched) and DIOR-R (public; in-house-trained detectors under Apache-2.0 mmrotate). The `rotcert` tool (numpy/scipy/shapely only; no torch/mmrotate in the certifier) and all frozen result records (DOTA pilot, DIOR-R Oriented R-CNN and RTMDet-R-l, the out-of-sample $R=20$ coverage, the naive regime audits, the frozen preregistered head-to-head record, the across-split analysis record, and the coverage-matched ablation record) accompany the paper; the CLI is golden-file-tested against these records. The `rotcert` tool and all frozen result records are archived at Zenodo (DOI: 10.5281/zenodo.21392293) with a `rotcert`-scoped public release of the development tree and records at <https://github.com/PeterPonyu/rotcert>, matching the scoped public-release pattern used for `PeterPonyu/asr-gate` and `PeterPonyu/inspect-gate`. The surrounding private `PeterPonyu/reliability-commons` monorepo is not published wholesale because it contains other manuscripts and submission materials. Provenance note: the frozen head-to-head record was written with a dirty working tree (uncommitted changes present when the preregistered family ran); the exact script and environment are stamped in the record’s own provenance metadata. Detector weights are frozen published checkpoints (DOTA) or in-house-trained (DIOR-R, redistribution subject to the DIOR-R license review; result-records-only if rehosting is barred). The Config-A extension records (four new G1 certification cells on HRSC2016-MS and the DOTA detector zoo, their $R=20$ recalibration, the G2 power-floor refusals, and the reversed registered-prediction verdict) accompany the paper as well, as do the Config-B extension records (two new G1 certification cells on DIOR-R — RoI Transformer, S2A-Net — their $R=20$ recalibration, G2 status, and the extended four-cell per-class conditional-coverage record).

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../figures-src/fig-seamsquare.pdf

Figure 2: Why the nonconformity score is a Wasserstein distance, not a coordinate gap (all GWD values computed with the released `rotcert` implementation). (a) A prediction at $\theta_{\text{pred}} = -88^\circ$ and the ground truth at $+88^\circ$ are 176° apart coordinate-wise but describe nearly the same box; the GWD assigns 0.13 px. (b) For a square box the long-edge angle is unidentifiable: the identical box at $\theta = 0^\circ$ and $\theta = 90^\circ$ is charged 90° by a coordinate-wise score and exactly 0 by GWD. (c) Sweeping the prediction angle with the ground truth fixed at $+88^\circ$: the raw coordinate gap spans 0 – 178° (and jumps at the wrap), while GWD stays small and smooth through the seam — a score whose spurious range is a hundred times its informative range forces vacuously wide conformal intervals. (d) The same argument measured, not argued: per-class certified radius ratio ($\text{GWD} \div \text{best baseline}$) across the 180 non-degenerate class cells of the family — GWD is smaller in 91/180 (with the three CI-excluding-zero cells marked in Figure 7d and none surviving Holm).



../figures-src/fig-thetarose.pdf

Figure 3: Per-class orientation distributions on the DIOR-R test split (frozen ground-truth record, angles canonicalized to the long-edge convention). Elongated classes (top row: ship, harbor, bridge; median aspect ratio 2.9–4.0) carry informative, multi-modal orientation structure — the certified orientation half-extent is tight there (2.9° – 11.1°). Near-square classes (bottom row; median aspect ratio 1.0–1.3) are isotropic by construction, so a predicted angle for them is noise rather than signal: the certificate correctly reports 90° (unconstrained) instead of certifying an unidentifiable quantity.

../figures-src/fig_showcase.pdf

Figure 4: The G1 certificate made concrete on four DIOR-R test scenes (S2A-Net Config-B cell). **Top row:** full scenes with matched predicted oriented bounding boxes (score ≥ 0.5). **Bottom row:** zoom crops of one certified object per scene, showing the predicted OBB (solid), ground truth (dashed white), the certified center-offset radius r_{center} (dashed circle), and the certified orientation half-extent $\Delta\theta_{\text{max}}$ (shaded wedge; omitted where orientation is unconstrained). The readouts are the same actionable quantities reported in Table 5; the IoU floor for each class is printed in the panel label. From the frozen actionable-certificate and matched-detection records.

figures-src/fig3_g2planning.pdf

Figure 5: The G2 power floor as a planning readout (post-freeze exploratory; closed-form inversion of the frozen recall records, no LTT re-run). **a:** recorded per-class calibration images n_{img} vs. the Bentkus power floor for the refused classes (log-log; the dashed line is “floor met”). **b:** per-cell regime split — certified, refused with a finite (reachable) floor, refused as unreachable ($\hat{r} \geq \beta$) — across all nine cells. **c:** classes certifiable with current data as the recall tolerance β relaxes; the frozen operating point $\beta=0.20$ is marked. **d:** distribution of the per-class β_{min} — the loosest tolerance at which each refused class would certify with the data it already has. From the frozen recall records.

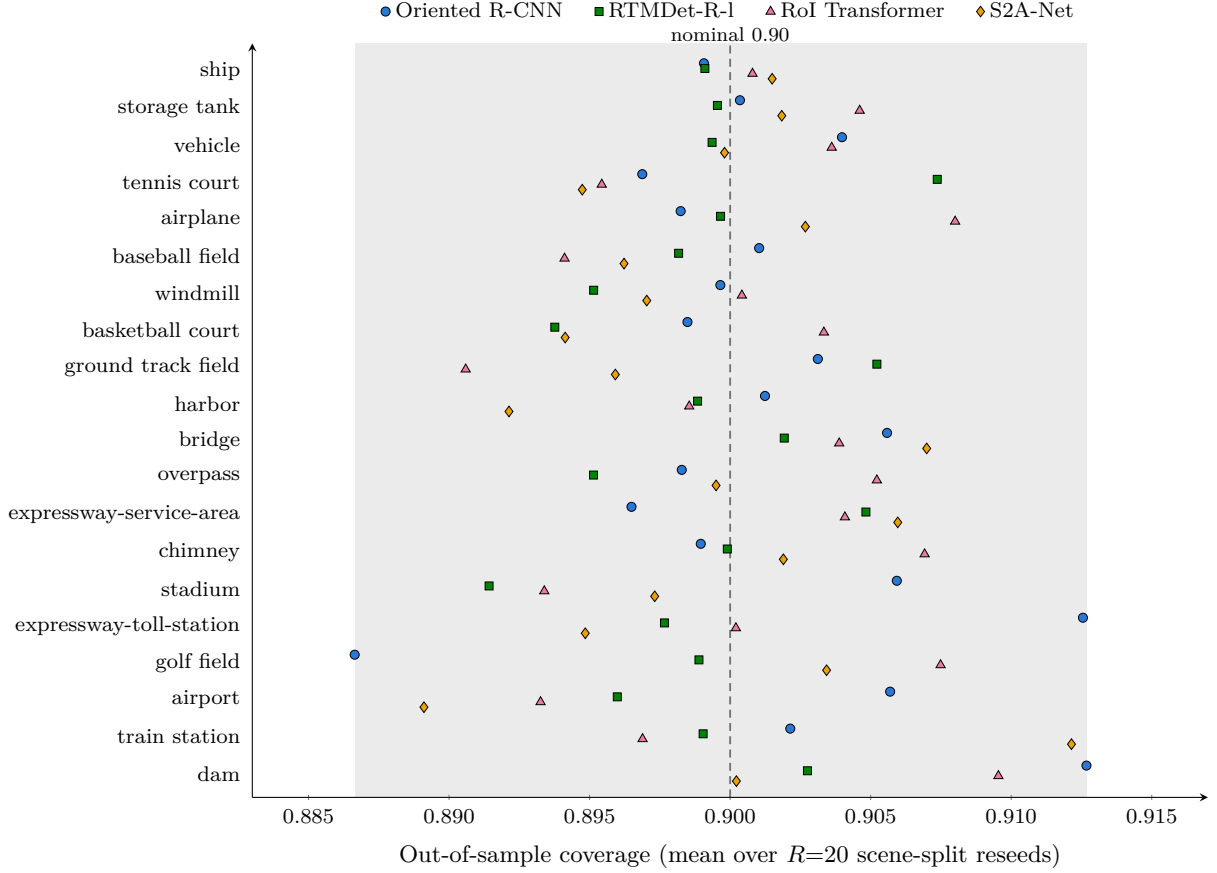


Figure 6: The four-cell per-class G1 record in detail ($\alpha = 0.10$, nominal 0.90, $R=20$ scene-split reseeds). **a:** per-class out-of-sample conditional coverage — all 80 class $\times$ cell points sit within the shaded ≈ 1.3 -point band around nominal. **b:** the reseed spread behind those means (s.d. median 0.020, max 0.053). **c:** certified GWD radius $\hat{q}$ vs. per-class calibration size: rare classes pay for nominal coverage with a wider region. **d:** the G1 certifiability floor $\alpha_{\min} = 1/(n_{\text{cal}}+1)$ — every class clears the frozen $\alpha=0.10$ by an order of magnitude; point colour marks the companion G2 recall verdict (certified vs. refused), separating the two certificates’ floors. From the frozen four-cell per-class conditional-coverage record.

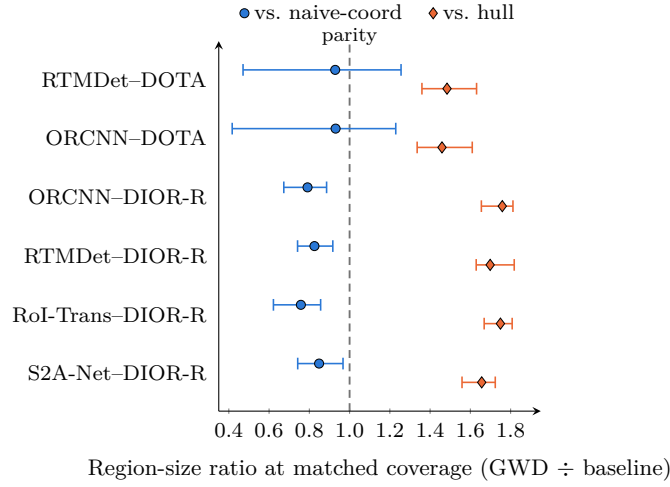


Figure 7: The coverage-matched efficiency test in full. **a:** per-cell region-size ratio ($\text{GWD} \div \text{baseline}$; <1 favors GWD) at realized coverage matched to the GWD certificate, $R=20$ mean with $[\text{min}, \text{max}]$; all four DIOR-R cells sit below parity against the naive baseline, both DOTA cells straddle it, every cell sits above the hull. **b:** all 240 per-split ratios — the verdict is not a mean artifact (104/240 splits below parity, cell-dependent). **c:** the coverage-matching mechanism itself: baseline α' per cell vs. the frozen $\alpha=0.10$ line GWD is held to. **d:** the 180 class-level non-degenerate cells sorted by area reduction against the best baseline; three carry CIs excluding zero and none survives Holm ($n_{\text{reject Holm}}=0$) — the powerless test reported as such, not amended. From the frozen coverage-matched and per-class records.